

JavaScript Modules

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What are Modules?

JavaScript modules allow you to break up your code into separate files.

This makes it easier to maintain the code-base.

Modules are imported from external files with the `import` statement.

Modules also rely on `type="module"` in the `<script>` tag.

Module Files

A module file exports functions or objects using the `export` keyword. The main file imports them using `import` :

math.js - Module file:

```
export const add = (a, b) => a + b;
export const multiply = (a, b) => a * b;
export const subtract = (a, b) => a - b;
export const divide = (a, b) => a / b;
```

Main HTML file:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Modules</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Module Files</h4>
<p id="demo"></p>

<script type="module">
import { add, multiply } from './math.js';

document.getElementById("demo").innerHTML =
  "5 + 3 = " + add(5, 3) + "<br>" +
  "5 * 3 = " + multiply(5, 3);
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

Export and Import

You can export individual functions and import them by name:

message.js - Module file:

```
export const greeting = "Hello";
export function welcome(name) {
  return greeting + ", " + name + "!";
}
export const farewell = (name) => "Goodbye, " + name + "!";
```

Main HTML file:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Modules</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Export and Import</h4>
<p id="demo"></p>

<script type="module">
import { greeting, welcome, farewell } from './message.js';

document.getElementById("demo").innerHTML =
  greeting + "<br>" +
  welcome("John") + "<br>" +
  farewell("Jane");
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Export Objects

You can export entire objects from a module:

person.js - Module file:

```
export const person = {
  firstName: "John",
  lastName: "Doe",
  age: 30,
  fullName: function() {
    return this.firstName + " " + this.lastName;
  }
};

export const cars = ["BMW", "Volvo", "Saab", "Ford"];
```

Main HTML file:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Modules</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Export Objects</h4>
<p id="demo"></p>

<script type="module">
import { person, cars } from './person.js';

document.getElementById("demo").innerHTML =
  "Person: " + person.fullName() + "<br>" +
  "Age: " + person.age + "<br><br>" +
  "Cars: " + cars.join(", ");
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Modules](#)